

1)PSO(1)

Particle swarm optimization (Best 5 genes)		
1	["DNAJB14", "SMARCD2", "ZBTB43", "MBD4", "COLEC10"]	RF Accuracy: 0.9353
2	["DNAJB14", "CTBS", "FGF23", "MSH3", "IL1RL1"]	GB Accuracy: 0.9118
3	["DNAJB14", "MSH3", "CTBS", "FGF23", "P2RY1"]	RF Accuracy: 0.9118
4	["DNAJB14", "COLEC10", "MSH3", "MBD4", "RECQL"]	GB Accuracy: 0.9059
5	["DNAJB14", "ZBTB43", "BMP5", "SMARCD2", "TOPORS"]	GB Accuracy: 0.9294
Average Accuracy: 0.9188 Common genes repeated more than 2 ["DNAJB14", "MSH3"]		

Particle swarm optimization (Best 10 genes)		
1	["DNAJB14", "MSH3", "TMEM100", "FGF23", "RECQL", "BAP1", "HS3ST3A1", "TESK1", "CAMK2G", "DOCK10"]	GB Accuracy: 0.9294
2	["DNAJB14", "SMARCD2", "COLEC10", "CTBS", "TCHH", "TMEM100", "BMP5", "MBD4", "BAP1", "RHOT2"]	RF Accuracy: 0.9353
3	["DNAJB14", "COLEC10", "SMARCD2", "BMP5", "CHST4", "RALA", "DOCK10", "BAP1", "SYNGR3", "FZD7"]	GB Accuracy: 0.9353
4	["DNAJB14", "CTBS", "TOPORS", "CHST4", "FGF23", "BMP5", "RECQL", "SMARCD2", "IL1RL1", "COLEC10"]	RF Accuracy: 0.9176
5	["DNAJB14", "WT1", "RALA", "COLEC10", "TCHH", "P2RY1", "RHOT2", "LIN37", "SLC2A9", "DOCK10"]	RF Accuracy: 0.9294
Average Accuracy: 0.9294 Common genes repeated more than 2 ["DNAJB14", "BAP1", "COLEC10", "BMP5", "CHST4"]		

Particle swarm optimization (Best 20 genes)		
1	["DNAJB14", "SMARCD2", "MSH3", "TMEM100", "FGF23", "RALA", "WT1", "TOPORS", "BAP1", "RECQL", "RHOT2", "WASL", "DOCK10", "HS3ST3A1", "HGF", "VIPR1", "LACTB2", "PDGFRA", "TESK1", "ADAMTS1"]	RF Accuracy: 0.9471
2	["DNAJB14", "CTBS", "FGF23", "RECQL", "PEG10", "BAP1", "WT1", "RND3", "ZBTB43", "RIMBP2", "DOCK10", "PDGFRA", "RHOT2", "WASL", "TESK1", "SYNGR3", "TMEM100", "CEP57", "P2RY1", "PTTG1"]	RF Accuracy: 0.9294
3	["DNAJB14", "FGF23", "ZBTB43", "TMEM100", "LACTB2", "MBD4", "RECQL", "IL1RL1", "BMP5", "ANKRD36", "PTTG1", "VIPR1", "SLC5A1", "PRPF4B", "PEG10", "HSPA1A", "PPP2R5C", "HS3ST3A1", "SAE1", "EP300"]	GB Accuracy: 0.9353
4	["DNAJB14", "MSH3", "CTBS", "TOPORS", "FGF23", "TMEM100", "CHST4", "ZBTB43", "DOCK10", "CAMK2G", "BAP1", "IL1RL1", "HGF", "SLC5A1", "FZD7", "ANKRD36", "FARP2", "ANKHD1-EIF4EBP3", "LIN37", "TESK1"]	RF Accuracy: 0.9353
5	["DNAJB14", "SMARCD2", "MBD4", "ZBTB43", "RALA", "COLEC10", "RECQL", "DIP2A", "FGF23", "CHST4", "NAT2", "TMEM100", "TCHH", "TERF1", "LIN37", "P2RY1", "FZD7", "WT1", "ZNF614", "USP48"]	Accuracy: 0.9294
Average Accuracy: 0.9353 Common genes repeated more than 2 ["DNAJB14", "FGF23", "BAP1", "TMEM100", "ZBTB43", "RECQL", "DOCK10", "RALA", "WT1", "TESK1"]		

2) PSO(2) and Ant Colony

Opt.	Particle swarm optimization	Ant Colony
Genes	Best 5 genes	
1	['MAST1', 'COLEC10', 'FASTK', 'SLC22A4', 'DNAJB14'] Best cross-validation: 0.9285500039466413 Test: 0.922077922077922	['DNAJB14', 'ING2', 'SAE1', 'RBM34', 'HS3ST3A1'] Best cross-validation: 0.9058489225668955 Test 0.9155844155844156
2	['RANBP2', 'RIMBP2', 'FASTK', 'ASF1B', 'DNAJB14'] Best cross-validation: 0.9091009550872208 Test: 0.922077922077922	['HS3ST3A1', 'BMP5', 'DNAJB14', 'LYVE1', 'SAE1'] Best cross-validation: 0.9123450943247297 Test 0.9090909090909091
3	'BMP5', 'HS3ST3A1', 'DNAJB14', 'CD47', 'TRIP13']	['CST8', 'ZFPL1', 'COPS3', 'ZBTB43', 'DNAJB14'] Best cross-validation: 0.8863683005762096

	Best cross-validation: 0.9220617254716236 Test: 0.9025974025974026	Test: 0.8701298701298701
4	['SLC12A3', 'AOAH', 'HGF', 'DNAJB14', 'PMVK'] Best cross-validation: 0.8993527508090615 Test: 0.8961038961038961	'HGF', 'SLC17A3', 'ASTN2', 'DNAJB14', 'ALPK3'] Best cross-validation: 0.8977583076801642 Test: 0.8961038961038961
5	['GALNT3', 'BAP1', 'LY6H', 'NEUROD2', 'DNAJB14'] Best cross-validation: 0.9188333727997474 Test: 0.922077922077922	Selected feature names: ['DNAJB14', 'RIMBP2', 'PELI2', 'CCDC40', 'KNTC1'] Best cross-validation 0.9058568158497119 Test 0.9285714285714286
	BEST: 0.922077922077922 AVG: 0.91298701298701 WORST: 0.8961038961038961	BEST: 0.9285714285714286 AVG: 0.9038961038961 WORST: 0.8701298701298701

Opt.	Particle swarm optimization	Ant Colony
Genes	Best 10 genes	
1	['SPINK1', 'CACNA1A', 'DNAJB14', 'FLCN', 'COL9A2', 'SMYD5', 'FZD7', 'RHOT2', 'SV2A', 'PDK3'] Best cross-validation: 0.920483068908359 Test: 0.9155844155844156	['SIX3', 'PODXL2', 'ATP5D', 'MTCH1', 'DAB2', 'CYP3A7', 'SLC12A3', 'IL1RL1', 'CAMK2G', 'DNAJB14'] Best cross-validation : 0.9204751756255427 Test = 0.8961038961038961
2	['BMP5', 'DNAJB14', 'ATP6V1G1', 'ADAMTS1', 'SLC17A3', 'COX6C', 'SAC3D1', 'ARMC4', 'ZAP70', 'SYNGR3'] Best cross-validation: 0.9269318809692951 Test: 0.9025974025974026	['COL5A2', 'DNAJB14', 'DDX58', 'SLC17A3', 'TANK', 'CILP', 'HS3ST3A1', 'PTN', 'RHOT2', 'TTC1'] Best cross-validation: 0.9253295445575814 Test = 0.948051948051948
3	['PELI2', 'HS3ST3A1', 'PDCD6IP', 'EDN2', 'NRIP1', 'EP300', 'HOXD1', 'TOP2A', 'DNAJB14', 'TMCC2'] Best cross-validation: 0.9415660273107586 Test: 0.9090909090909091	['FZD7', 'SLC2A9', 'CD47', 'TMCC2', 'TMEM100', 'ARL2', 'PAF1', 'ARMC1', 'DNAJB14', 'RCE1'] Best cross-validation: 0.9253295445575814 Test = 0.9090909090909091

4	['BMP5', 'NOL12', 'MAST1', 'SLC34A2', 'SCN4A', 'RCE1', 'TRIP13', 'ZNF551', 'COMT', 'DNAJB14'] Best cross-validation: 0.936695871813087 Test: 0.9090909090909091	['IGFALS', 'DNAJB14', 'NLRP3', 'VSNL1', 'ZNF446', 'ZNF148', 'CALCA', 'CHST4', 'CDKN2C', 'FASTK'] Best cross-validation 0.9302075933380692 Test 0.9090909090909091
5	['BMP5', 'PALM', 'NAT2', 'PRAMEF10', 'SAC3D1', 'DNAJB14', 'PRSS8', 'ANKHD1-EIF4EBP3', 'IRX5', 'CAMK2G'] Best cross-validation: 0.9269950272318258 Test 0.9090909090909091	['DNAJB14', 'CD47', 'VIPR1', 'HOXD1', 'SLC1A1', 'GPR89B', 'AGGF1', 'RAB3A', 'MOSPD1', 'CEP57'] Best cross-validation 0.9285500039466413 Test 0.8961038961038961
	BEST: 0.9155844155844156 AVG: 0.9090909090909091 WORST: 0.9025974025974026	BEST: 0.948051948051948 AVG: 0.91168831168831 WORST: 0.8961038961038961

Opt.	Particle swarm optimization	Ant Colony
Genes	Best 20 genes	
1	['IAPP', 'RYSR2', 'PTPRM', 'USP48', 'PRPF4B', 'SLAMF7', 'FAM53C', 'RIN3', 'POMZP3', 'ATP6V1B2', 'CDH19', 'DNAJB14', 'LYRM2', 'CNPY3', 'SLC17A3', 'PELI2', 'COX6C', 'HTR1B', 'TERF1', 'CASR'] Best cross-validation: 0.9431920435709212 Test: 0.9415584415584416	['RAB3A', 'PDK3', 'IGFALS', 'CNTD2', 'SLC12A3', 'RNASE6', 'NEFH', 'MYH11', 'MAST1', 'LIN37', 'MYOM2', 'GABARAPL2', 'FASTK', 'S100A8', 'CILP', 'MAP4K3', 'PRKAR1B', 'DAB2', 'DNAJB14', 'NNMT'] Best cross-validation: 0.9383534612045149 Test 0.935064935064935
2	['PAF1', 'KCNN3', 'ERN1', 'ATP6V1B2', 'HSPA14', 'CLN3', 'SLAMF7', 'ATP6V1G1', 'IGFALS', 'HMGN4', 'VSNL1', 'CSH2', 'SLC17A3', 'SLC5A1', 'FASTK', 'NRIP1', 'PTTG1', 'DNAJB14', 'CWF19L1', 'ZNF551'] Best cross-validation: 0.956200173652222 Test: 0.9415584415584416	['KLRG1', 'RAB3D', 'GIP', 'NAT2', 'NDUFB4', 'ZBTB43', 'MDFI', 'MOSPD1', 'OTUD3', 'ERN1', 'HTR1B', 'TOPORS', 'PLVAP', 'MYH11', 'CCDC40', 'GABRB3', 'DNAJB14', 'TCHH', 'ADAMTSL2', 'LTA4H'] Best cross-validation: 0.935085642118557 Test 0.9155844155844156
3	['DNAJB14', 'IL1RL1', 'CTSC', 'MUC13', 'SAC3D1', 'POLR3C', 'PLA2G3', 'NPC1L1', 'CILP', 'NGFR', 'ZBTB39', 'MYL6', 'ERN1',	['SERHL2', 'TMEM45A', 'SLC28A2', 'ATXN2', 'PRPF4B', 'TFE3', 'PDGFRA', 'CCDC40', 'TOM1', 'TRIP13', 'DNAJB14', 'FSTL4', 'ADAMTSL2', 'CRADD', 'TBX5', 'CYP1A2', 'PHACTR2', 'TMSB10', 'TSPAN32', 'MAST1']

	'LRR31', 'DSPP', 'CHST4', 'ZNF148', 'TRIP13', 'P2RY1', 'HJURP'] Best cross-validation: 0.9545504775436103 Test: 0.9155844155844156	Best cross-validation: 0.9415660273107586 Test: 0.922077922077922
4	'MYOM2', 'TIA1', 'DNAJB14', 'ERN1', 'TM6SF1', 'SLC5A1', 'CYP1A1', 'GPR89B', 'CDKN2C', 'RACGAP1', 'KNTC1', 'SPANXA1', 'TRIP13', 'ASF1B', 'PRAMEF10', 'PGM1', 'NDUFA4L2', 'TSPAN32', 'PDGFRA', 'PELI2'] Best cross-validation: 0.9415818138763911 Test: 0.935064935064935	['HCLS1', 'SMARCD2', 'CHST4', 'PTPRM', 'PLEKHH3', 'RACGAP1', 'MBD4', 'ACTL6A', 'ELMO1', 'HS3ST3A1', 'DNAJB14', 'AGGF1', 'HK1', 'IGFALS', 'RAB3A', 'PDGFRA', 'TIA1', 'CNPY3', 'DYNLT3', 'PELI2'] Best cross-validation: 0.9334517325755782 Test: 0.922077922077922
5	['DNAJB14', 'PPP2R2B', 'PSMC1', 'ALOX15', 'FNBP1', 'MPP1', 'GGT1', 'RPS3', 'TMEM106C', 'NEUROD2', 'PMVK', 'HIST1H2AM', 'TSPAN5', 'COPS3', 'ZFPL1', 'CYP11A1', 'CD34', 'VIPR1', 'CCDC40', 'PCBP3'] Best cross-validation 0.9367116583787197 Test: 0.9415584415584416	['MCFD2', 'ARMC4', 'NEFH', 'BET1', 'VEZF1', 'PA2G4', 'CASR', 'DNAJB14', 'FNTA', 'PEG10', 'CAMK2G', 'IGLL1', 'RAB3D', 'AP3B2', 'PQLC2', 'CALCA', 'FASTK', 'ZBTB39', 'SNCAIP', 'EPHA3'] Best cross-validation: 0.9350698555529244 Test 0.9155844155844156
	BEST: 0.9415584415584416 AVG: 0.93506493506494 WORST: 0.9155844155844156	BEST: 0.935064935064935 AVG: 0.92207792207792 WORST: 0.9155844155844156

4) Genetic Algorithm

Run Number	Number of Genes	Selected Genes (Gene Names)	Accuracy (%)
1	5	["NDUFA4L2", "HS3ST3A1", "TIPRL", "DNAJB14", "RIC3"]	Random Forest: 0.9176 Gradient Boosting: 0.9235 SVC: 0.8647 MLP Classifier: 0.8706
2	5	["COLEC10", "CDH19", "FASTK", "DNAJB14", "RIC3"]	Random Forest: 0.9412 Gradient Boosting: 0.9353 SVC: 0.8882 MLP Classifier: 0.8824
3	5	["TBX5", "HIST1H2BJ", "ASF1B", "DNAJB14", "VIPR1"]	Random Forest: 0.9176 Gradient Boosting: 0.9118 SVC: 0.9118 MLP Classifier: 0.8824

Run Number	Number of Genes	Selected Genes (Gene Names)	Accuracy (%)
1	10	["NOL8", "SYNGR3", "ZNF362", "DSC2", "TMEM106C", "PRSS22", "FERMT2", "BCAS4", "PDE6G", "DNAJB14"]	Random Forest : 0.9353 Gradient Boosting : 0.9353 SVC : 0.9118 MLP Classifier : 0.8824
2	10	["PRAMEF10", "HJURP", "PRSS8", "KLRG1", "HIST1H2AM", "RAB3IL1", "MPZL1", "ARID3B", "COLEC10", "DNAJB14"]	Random Forest : 0.9353 Gradient Boosting : 0.9353 SVC : 0.8882 MLP Classifier: 0.8588
3	10	["TMEM106C", "INTS8", "HJURP", "GNL3", "VEZF1", "DSPP", "CAMK2G", "IPPK", "DNAJB14"]	Random Forest: 0.9176 Gradient Boosting: 0.9059 SVC: 0.8529 MLP Classifier: 0.8235

Run Number	Number of Genes	Selected Genes (Gene Names)	Accuracy (%)
1	20	["PALM", "OTUD3", "ABCC3", "ACTL6A", "USP48", "PLEKHG3", "RBM34", "MCTP1", "SRP19", "KLRG1", "ATXN2", "PRNP", "DGKH", "CILP", "DNASE1L2", "TEK", "BAP1", "S100A8", "CCDC40", "DNAJB14"]	Random Forest: 0.9353 Gradient Boosting: 0.9471 SVC : 0.9294 MLP Classifier: 0.8353
2	20	["PRKAR1B", "YBX2", "PLEKHG3", "OR12D2", "CNTD2", "INTS8", "RHCE", "LACTB2", "C9", "COPS3", "RECQL", "CXCR6", "CTSC", "VGLL4", "ZNF614", "CEP57", "CILP", "ALPK3", "CCDC40", "DNAJB14"]	Random Forest : 0.9412 Gradient Boosting :0.9353 SVC : 0.8941 MLP Classifier: 0.8059
3	20	["FNTA", "EXOC3", "NOL8", "PRPF4B", "NCAPG", "ALOX15", "TCHH", "UFC1", "RYSR2", "GPR89B", "SPANXA1", "HTR1B", "SAE1", "TRIM28", "CILP", "RIC3", "COLEC10", "RASSF1", "SLC2A9", "DNAJB14"]	Random Forest: 0.9294 Gradient Boosting: 0.9647 SVC : 0.8882 MLP Classifier: 0.8294

Grey Wolf

Opt.	GWO
Genes	5
1	Iteration 20, Best Fitness: 0.935064935064935 Selected Features: ['KLRB1', 'TIMM10', 'TMEM106C', 'RASSF1', 'DNAJB14']
2	Iteration 20, Best Fitness: 0.9090909090909091 Selected Features: ['PALM', 'WDR91', 'NLRP2', 'DDX49', 'DNAJB14']
3	Iteration 20, Best Fitness: 0.9285714285714286 Selected Features: ['IGF2BP3', 'DIP2A', 'LYVE1', 'STK16', 'DNAJB14']
4	Iteration 20, Best Fitness: 0.9155844155844156 Selected Features: ['SCNN1D', 'TESK1', 'COX6C', 'CTBS', 'DNAJB14']
5	Iteration 20, Best Fitness: 0.9285714285714286 Selected Features: ['TIA1', 'ELMO1', 'TSPAN5', 'COLEC10', 'DNAJB14']
	BEST: 0.935064935064935 AVG: 0.92337662337662 WORST: 0.9090909090909091

Opt.	GWO
Genes	10
1	Iteration 20, Best Fitness: 0.9545454545454546 Selected Features: ['FOXG1', 'CADPS', 'IGF2BP3', 'ABCD1', 'ADAMTS1', 'VGLL4', 'MAGEA1', 'BMP8B', 'COLEC10', 'DNAJB14']
2	Iteration 20, Best Fitness: 0.961038961038961 Selected Features: ['RANBP2', 'SLC1A1', 'ZXDA', 'LIN37', 'BNIP3L', 'SPANXA1', 'TRIM28', 'CLN3', 'GPR21', 'DNAJB14']
3	Iteration 20, Best Fitness: 0.9415584415584416 Selected Features: ['LMAN1', 'IRF9', 'ARHGEF1', 'DSC2', 'HCLS1', 'EPHA3', 'ASTN2', 'PDK3', 'ABCG5', 'DNAJB14']
4	Iteration 20, Best Fitness: 0.935064935064935 Selected Features: ['FNTA', 'CADPS', 'OTUD3', 'NOL8', 'TOPORS', 'PGBD5', 'RYSR2', 'MOSPD1', 'COLEC10', 'DNAJB14']
5	Iteration 20, Best Fitness: 0.948051948051948

	Selected Features: ['WASL', 'TBX5', 'NOL8', 'ZZZ3', 'ZNF446', 'ADAMTS1', 'BATF', 'IGFALS', 'TTC1', 'DNAJB14']
	BEST: 0.961038961038961 AVG: 0.94805194805195 WORST: 0.935064935064935

Opt.	GWO
Genes	20
1	Iteration 20, Best Fitness: 0.974025974025974 Selected Features: ['HNRNPAB', 'WDR43', 'RNASET2', 'PRPF4B', 'SYNGR3', 'PA2G4', 'CLN8', 'STK3', 'TCHH', 'NFATC1', 'BNIP3L', 'LYVE1', 'ARL2', 'APOD', 'MAGEA1', 'CILP', 'MLC1', 'ARID3B', 'CTBS', 'DNAJB14']
2	Iteration 20, Best Fitness: 0.974025974025974 Selected Features: ['FNTA', 'PRKAB2', 'HK1', 'IGF2BP3', 'AQP6', 'WDR43', 'RNASET2', 'WDR91', 'ARTN', 'MMRN1', 'TANK', 'PRSS22', 'YPEL1', 'PMVK', 'GABRB3', 'HS3ST3A1', 'PDK3', 'LEPR', 'PDGFRA', 'DNAJB14']
3	Iteration 20, Best Fitness: 0.987012987012987 Selected Features: ['HK1', 'KLRB1', 'YBX2', 'MAL', 'RAB3A', 'RBM34', 'SULT1C2', 'SULT1E1', 'TMEM106C', 'ATXN2L', 'C9', 'COPS3', 'EPB41L4A', 'MFN1', 'NUBPL', 'PRNP', 'DKK1', 'TMEM100', 'CCDC40', 'DNAJB14']
4	Iteration 20, Best Fitness: 0.961038961038961 Selected Features: ['PRKAB2', 'HK1', 'WDR43', 'GABARAPL2', 'SLC1A1', 'CRKL', 'BET1', 'HSPA1A', 'GATM', 'COX6C', 'GCLM', 'SLC12A3', 'BCAS4', 'ARL2', 'AKR1B10', 'TOP2A', 'CEP57', 'FGF23', 'COMT', 'DNAJB14']
5	Iteration 20, Best Fitness: 0.974025974025974 Selected Features: ['FOXG1', 'KLRB1', 'EXOC3', 'AQP6', 'MCFD2', 'ZXDA', 'LMAN1L', 'TMEM106C', 'PRSS22', 'GFI1', 'CXCR6', 'FSTL4', 'RIMBP2', 'EPHA3', 'PPID', 'CYP11A1', 'MTCH1', 'LECT2', 'GPR21', 'DNAJB14']
	BEST: 0.987012987012987 AVG: 0.97402597402597 WORST: 0.961038961038961